

ESSEY ARAYA

Curriculum Vitae

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EDUCATION

Berea College

B.S. in Engineering Physics and Computer Science

Berea, KY

Expected Dec 2027

- GPA: **3.93/4.00**
- Minor in Mathematics
- Relevant Coursework: Machine Learning, Numerical Analysis, Differential Equations, Calculus III, Discrete Maths, Data Structures & Algorithms, Design & Analysis of Algorithms, Embedded Systems, Electromagnetic Theory, Classical Mechanics, Thermodynamics, Power and Energy Systems, Advanced General Lab

RESEARCH INTERESTS

Scientific ML; physics-informed modeling; Bayesian system identification; uncertainty quantification; scientific computing and software development; applications in energy, sustainability, materials, and autonomous systems.

RESEARCH EXPERIENCE

Machine Learning Research Intern

Vanderbilt University School of Engineering

May 2026 – Aug 2026

Nashville, TN

- Built and integrated GP-PHS into NeuroMANCE using Python, PyTorch, and GPyTorch, enabling researchers to model robotic systems with uncertainty-aware machine learning
- Designed and implemented 9 modular components for data pipelines, matrix operations, and numerical computation using NumPy, Pandas, and SciPy, providing reusable infrastructure for ML training
- Reduced prediction error by 34% compared to standard Gaussian processes by embedding physics
- Held weekly Agile review meetings with a 6-member team, incorporating feedback to iteratively develop comprehensive API documentation and 5 tutorial notebooks
- Created 250+ unit, integration, and end-to-end tests using Pytest to improve reliability

Materials Research Intern

Berea College

May 2025 – Jul 2025

Berea, KY

- Designed and fabricated a lightweight composite battery enclosure for Berea College's first solar car using Ansys ACP composite simulator and wet layup, reducing projected manufacturing cost by 50%
- Eliminated 80% of testing costs by simulating FEA tests in Ansys and validated designs using a UTM
- Created female molds from 3D-printed plugs, cutting fabrication time by 20+ hours and avoiding CNC costs
- Maintained 10+ lab tools and organized an inventory of 50+ prototyping materials to streamline workflow

PUBLICATIONS

GP-PHS | Abstract submitted to IFAC SYSID 2027

Sep 2026

- Araya, E., and Beckers, T. "GP-PHS: A Toolbox for Physics-Informed Bayesian Modeling of Port-Hamiltonian Systems."

ENGINEERING EXPERIENCE

Machine Learning Engineer Intern / AI Studio Fellow

NASA × Break Through Tech

Sep 2026 – Present

Remote / New York, NY

- Developed a RAG and agentic AI system to automate software requirements auditing against NASA engineering standards, reducing compliance review time from weeks to minutes.

Mechanical Team Lead

Berea Engineering and Racing, Berea College

Aug 2025 – May 2026

Berea, KY

- Led a 3-member team to optimize the chassis, reducing weight by 32% while meeting safety targets
- Conducted 50+ FEA simulations using Altair Inspire and validated critical cases through hand calculations

TEACHING EXPERIENCE

Teaching Assistant (Solar Vehicle Aerodynamics)

Berea College

Aug 2026 – Present

Berea, KY

- Mentored a student in understanding and applying computational fluid dynamics for solar-powered cars

Physics Teaching Assistant

Berea College

Aug 2025 – May 2026

Berea, KY

- Tutored 15+ students in introductory physics by breaking down complex concepts into clear explanations
- Held review sessions for students struggling with course material, increasing exam passing rates by 15%

PROJECTS

Thermal Systems Modeling

Python, NumPy, SciPy, Finite Differences

[View Project](#)

- Collaborated in a 6-person team to build first- and second-order ODE models in Python to predict passenger thermal comfort from 299-point temperature datasets from a sensor
- Used SciPy, finite differences, and parameter sensitivity studies to identify how heat transfer and thermal inertia drive temperature swings while evaluating numerical stability and accuracy

Drone Vertical Dynamics Simulation

C++, Python, GitHub Actions (CI/CD), Pandas, Matplotlib

[View Project](#)

- Built a C++ physics simulator modeling gravity, thrust, and drag, producing velocity and height data
- Implemented CI/CD with GitHub Actions to compile, run, validate, and deploy outputs automatically

Electromyography Signal Acquisition

OpAmps (LF353), Oscilloscope, Multistage Circuit

[View Project](#)

- Designed and built a multi-stage EMG amplifier system to acquire tiny bioelectric signals from the muscle
- Characterized and validated system gain and bandwidth across varying muscle loads through testing

Lumen – Team Lead

Next.js 16, TypeScript, PostgreSQL, Vercel AI SDK

[View Project](#)

- Built a greedy prerequisite-graph solver in TypeScript to automate 4-year scheduling for 1,500+ students
- Developed a course-ranking pipeline that translated students' career goals into 3 personalized degree pathways, helping students align coursework with their desired careers

SafeSource – Hackathon

Python, FastAPI, Gemini Vision API, Pydantic v2, Next.js 15

[View Project](#)

- Built a platform that analyzed environmental data to help users identify health risks by quantifying PFAS
- Developed a Google Gemini Vision + Pydantic pipeline to estimate PFAS exposure from grocery receipt images, helping users identify high-risk products in everyday purchases

LEADERSHIP & CAMPUS INVOLVEMENT

President

Berea Engineering and Racing

Mar 2026 – Present

Berea, KY

- Led weekly meetings and organized hackathons and design competitions, reaching 100+ participants

Vice President

Berea Engineering and Racing

May 2025 – Mar 2026

Berea, KY

- Led 5+ engineering workshops and collaborative builds to mentor members in CAD, FEA, and prototyping

HONORS & AWARDS

- Vanderbilt University summer research stipend award: \$8,000 (2026)
- Berea College URCPP stipend award: \$5,000 (2025)
- 4-year tuition promise scholarship (\$285,000+ USD)
- Dean's List: All Semesters

CERTIFICATIONS

Machine Learning Foundations Cornell University	Aug 2026
Introduction to Programming with MATLAB Vanderbilt University	Aug 2025
Mechanics of Materials I: Fundamentals of Stress & Strain and Axial Loading Georgia Institute of Technology	Jun 2025
Supervised Machine Learning: Regression and Classification DeepLearning.AI / Stanford	Apr 2023
HTML, CSS, and JavaScript for Web Developers Johns Hopkins University	Sep 2022
IBM Data Analyst Specialization IBM; Professional Certificate, ACE credit recommendation	Sep 2022
Git for Developers Using Github Coursera	Aug 2022
Google IT Support Specialization Google; Professional Certificate, ACE credit recommendation	Jul 2022
Web Design for Everybody: Basics of Web Development & Coding Specialization University of Michigan	Dec 2021

TECHNICAL SKILLS

Languages: C + +, Python, MATLAB, TypeScript, JavaScript, SQL

ML & Developer Tools: PyTorch, NumPy, SciPy, Pandas, Matplotlib, Git, Linux, Pytest, GitHub Actions

Scientific ML: GPyTorch, NeuroMANCER, Gaussian processes, physics-informed modeling, Bayesian inference, uncertainty quantification

Simulation & Engineering: Ansys Mechanical, Ansys ACP, Ansys Fluent, Altair Inspire, Autodesk Inventor, FEA, CFD, composite fabrication, wet layup, 3D printing, Universal Testing Machine

Web & Data: React, Next.js, FastAPI, PostgreSQL, HTML/CSS/SCSS, PostCSS, Netlify, Vercel AI SDK, Pydantic, Excel, IBM Cognos

Instrumentation: Microcontrollers, stepper motors, oscilloscopes, operational amplifiers, signal acquisition